

Informedness of the gradient-informed single-site proposal

Why it scales as $\Delta t/\sqrt{T}$, how Δt is chosen,
and what the $\Delta t = 460$ vs. $\Delta t = 63$ runs showed

my_extendedSampling / softmax

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Abstract

The sampler in `classes/rate_sampler.py` proposes one substitution per Monte Carlo move by reading off which amino acid a gradient-informed Gaussian displacement “points at”. How reliably that displacement follows the gradient’s own preference—its *informedness*—is controlled by Δt , but *not* by Δt alone: the governing combination is $\Delta t/\sqrt{MT}$. This note derives that scaling from the proposal construction, records the measured informedness curves for both reference sequences, explains why Δt is chosen at a deliberately *partial* informedness of $p_{\text{match}} \approx 0.15$ rather than as large as possible, and reports what the paired $\Delta t = 460$ / $\Delta t = 63$ Boltzmann-validation runs on `zero_polymer` ($L = 566$) established. The short version of the last point: both runs reproduce the exactly-enumerated Boltzmann distribution, so the Δt error that motivated the pair cost test *coverage*, not correctness. A subsequent doubling of the run length confirms the residual discrepancy decays as finite-sample noise must, in two independent functionals.

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1 Scope and sources

Everything below is either derived from the code as it stands or measured in a run whose output file is committed. Source of truth for each:

- Proposal construction: `classes/rate_sampler.py`, method `_step`.
- Pointing probability: `utils/proposals.py`, function `log_pointing_prob`.
- Informedness curves: `tests/test_informedness_vs_dt.py`; outputs `protein_g/informedness_vs_dt_results.txt` and `zero_polymer/informedness_vs_dt_results.txt`.
- Boltzmann validation: `tests/validate_boltzmann_toy.py`; outputs `zero_polymer/validate_boltzmann_results_dt460.txt` and `..._dt63.txt`.
- Chronology and the mistake that prompted this note: `softmax/DEVLOG.txt`, entries dated 2026-10-02 and 2026-10-03.

Quantities that are *estimated* rather than measured are flagged as such at the point of use; there are two (the effective sample size in §6.4, and the p_{match} interpolations in Table 2).

2 The proposal, and what “informedness” means

2.1 One move

A sequence of length L is held as per-site logits over the $K = 25$ -letter vocabulary `SEQUENCE_USED_VOCAB`, mean-centred along the class axis to gauge-fix the translation invariance of the softmax. One move:

1. Pick one site s uniformly from $\{0, \dots, L-1\}$. Uniform and state-independent, so it cancels exactly from the Metropolis–Hastings ratio and needs no correction term.
2. At that site only, draw a displacement $\delta x \in \mathbb{R}^K$ (§2.2) and propose the substitution it points at, *excluding* the site’s current residue and the five non-canonical codes `X, B, U, Z, O`. That leaves

$$K_c = 25 - 5 - 1 = 19$$

competitors. (K_c is 20 in the one edge case where the site already sits on a non-canonical code, since then there is no current residue to remove from the canonical set.)

3. Accept or reject with an exact Metropolis–Hastings correction using the true recomputed energy of the candidate and the true reverse proposal probability, so the target is $\exp(-U/T)$ exactly rather than to first order.

Note that step 2 always proposes a genuine change: “staying” is decided by rejection in step 3, never by the proposal.

2.2 The displacement

The displacement is one HMC leapfrog half-step with fresh momentum, which for a single step is the same object as a MALA/Langevin step:

$$\delta x = \frac{\Delta t p}{M} - \frac{\Delta t^2}{2M} \nabla U, \quad p \sim \mathcal{N}(0, MT I), \quad (1)$$

with $\nabla U = \partial U / \partial(\text{logits at } s)$ obtained by relaxing *only* site s through $\text{softmax}(\cdot / T_{\text{sftm}})$ while every other site is held at its exact one-hot value. Equation (1) makes δx Gaussian, and this is the pair of quantities everything downstream depends on:

$$\delta x \sim \mathcal{N}(\mu, \sigma^2 I), \quad \boxed{\mu = -\frac{\Delta t^2}{2M} \nabla U, \quad \sigma = \Delta t \sqrt{\frac{T}{M}}} \quad (2)$$

So the drift grows as Δt^2 while the noise grows only as Δt . That asymmetry is the entire mechanism by which Δt buys informedness, and §4 is just this observation carried through carefully.

2.3 Informedness, defined

Write j^* for the competitor the gradient alone would pick,

$$j^* = \arg \max_{j \in \text{competitors}} \mu_j = \arg \min_{j \in \text{competitors}} (\nabla U)_j, \quad (3)$$

the second equality because $\mu = -\Delta t^2 / (2M) \cdot \nabla U$ and $\Delta t^2 / (2M)$ is a *positive scalar*. This gives the first of two facts that the diagnostic `test_informedness_vs_dt.py` was built to separate:

- (A) **The gradient’s choice is independent of Δt .** Scaling μ by a positive scalar cannot change its $\arg \max$. So *which* residue the gradient prefers, and therefore how good that preference is, does not depend on Δt at all. Δt cannot improve the gradient’s taste.
- (B) **How often the realised draw obeys that choice does depend on Δt .** The actual proposal is the $\arg \max$ of δx , which is μ plus noise. Define

$$p_{\text{match}} = \mathbb{P} \left(\arg \max_j \delta x_j = j^* \right), \quad (4)$$

the probability that one realised move follows the gradient’s own pick. This is what Δt controls, and it is what “informedness” means throughout this note.

Conflating (A) and (B) was the original error that this decomposition fixed: an earlier diagnostic swept Δt looking for improvements in proposal *quality*, which by (A) cannot exist. The two limits of (B) are

$$p_{\text{match}} \longrightarrow \frac{1}{K_c} = \frac{1}{19} \approx 0.0526 \quad (\Delta t \rightarrow 0), \quad p_{\text{match}} \longrightarrow 1 \quad (\Delta t \rightarrow \infty), \quad (5)$$

i.e. a uniform pick among the 19 competitors at small Δt , and a deterministic, greedy gradient-descent pick at large Δt . Note the small- Δt limit is *uniform*, not “stay put”: nothing in (1) references the current residue’s identity, which is why an earlier design that tried to make small Δt mean “few sites change” could not work and was abandoned.

3 The pointing probability $a_{i \rightarrow j}$

3.1 Closed form

p_{match} , and more generally the proposal probability needed for the Metropolis–Hastings ratio, is the probability that a particular component of a Gaussian vector is the largest. “Pointing at e_j ” means $\delta x_j > \delta x_k$ for every competitor k , i.e. every other component falls below δx_j . Conditioning on the value of δx_j and using independence across k :

$$a_{i \rightarrow j} = \mathbb{P}(\delta x_j > \delta x_k \ \forall k \neq i, j) = \int_{-\infty}^{\infty} d\delta x_j f_{\mu_j, \sigma_j}(\delta x_j) \prod_{k \neq i, j} \int_{-\infty}^{\delta x_j} d\delta x_k f_{\mu_k, \sigma_k}(\delta x_k), \quad (6)$$

where $f_{\mu, \sigma}$ is the Gaussian density. The inner integrals are Gaussian cumulatives, so this is $\int d\delta x_j f_{\mu_j, \sigma_j}(\delta x_j) \prod_k \Phi_{\mu_k, \sigma_k}(\delta x_j)$. Standardising the outer variable, $\delta x_j = \mu_j + \sigma_j u$:

$$a_{i \rightarrow j} = \int_{-\infty}^{\infty} \phi(u) \prod_{k \neq i, j} \Phi(\alpha_{kj} + \beta_{kj} u) du, \quad \alpha_{kj} = \frac{\mu_j - \mu_k}{\sigma_k}, \quad \beta_{kj} = \frac{\sigma_j}{\sigma_k}. \quad (7)$$

with ϕ, Φ the standard normal density and CDF. Note that α_{kj} is divided by σ_k — the standard deviation of the competitor being compared against — not by σ_j .

The specialisation used here. In this sampler σ is a global scalar: $\delta x = \Delta t p/M - \frac{\Delta t^2}{2M} \nabla U$ with isotropic momentum $p \sim \mathcal{N}(0, MTI)$, so every component has the same variance $\Delta t^2 T/M$ and

$$\beta_{kj} = \frac{\sigma_j}{\sigma_k} = 1 \quad \text{identically,} \quad \alpha_{kj} = \frac{\mu_j - \mu_k}{\sigma}, \quad (8)$$

which is what `utils/proposals.py` implements (it takes σ as a scalar argument). Dropping β is therefore correct *here* but not in general: verified numerically against brute-force Monte Carlo, (7) reproduces the true pointing probabilities to within Monte Carlo error in both cases, whereas using a single σ when the σ_k genuinely differ is wrong by up to 0.135 in absolute probability on a $K = 6$ test. Any future variant with a per-class or per-site step size reinstates β .

With constant σ this is a Thurstone/probit discrete-choice model—the probit analogue of the softmax-based proposal used by Gibbs-With-Gradients. It is an exact probability, not merely something proportional to one: summing (7) over all admissible j at a fixed site gives 1, which is checked numerically in `tests/test_proposals.py`.

3.2 How it is evaluated

`log_pointing_prob` evaluates (7) by Gauss–Hermite quadrature entirely in log space, using $n_{\text{quad}} = 40$ nodes:

$$\log a_{i \rightarrow j} = \log \sum_q \exp \left[\log w_q + \sum_{k \neq j} \log \Phi(\alpha_{kj} + u_q) \right], \quad (9)$$

computed with `torch.special.log_ndtr` and `logsumexp` so that no intermediate underflows even when α is large. The current residue is removed by *slicing* it out of the tensor being maximised over, not by adding a large negative penalty—a deliberate change from an earlier version, since a finite penalty could in principle be approached by Δt^2 -scaled gradients at large Δt , whereas a sliced-out class is structurally absent.

3.3 The independence assumption is exact here, not an approximation

Equation (7) treats the K components of δx as independent, and the project has long recorded this as “a deliberate, small, controlled approximation” — the reasoning being that mean-centring the momenta and the displacement induces the covariance $\sigma^2(I - J/K)$ rather than $\sigma^2 I$, a pairwise correlation of $-1/(K - 1)$. That description is **wrong for the quantity actually being computed**, and the code is more correct than its own comment claimed.

The sampler centres twice — once inside `_extract_momenta` ($p \leftarrow p - \bar{p}$) and once on the displacement ($\delta x \leftarrow \delta x - \bar{\delta x}$). Both subtract the *same scalar from every component*, and $\arg \max$ is invariant under a common shift. The realised proposal is therefore the $\arg \max$ of the *uncentred* displacement, whose components are genuinely independent $\mathcal{N}(\mu_k, \sigma^2)$. The induced correlation never reaches the functional being evaluated. Slicing to the 19 competitors afterwards does not disturb this, since the same scalar was subtracted from those components too.

Verified by simulating the exact pipeline — centred momentum, centred displacement, $\arg \max$ over the competitor slice, 2×10^6 draws:

$\max \text{empirical} - \text{formula}(\sigma) $	0.00033
$\max \text{empirical} - \text{formula}(\sigma\sqrt{1 - 1/K}) $	0.00252
Monte Carlo standard error	0.00026

The formula with the *uncentred* σ agrees within Monte Carlo error. The “corrected” marginal $\sigma\sqrt{1 - 1/K}$ — the natural thing to reach for once one notices the centring — is wrong by about ten standard errors. Drawing uncentred momenta directly reproduces the centred pipeline’s $\arg \max$ frequencies to 0.00036, confirming the shift-invariance argument rather than merely asserting it.

So $a_{i \rightarrow j}$ as computed is exact, up to quadrature error, which is separately verified: Gauss–Hermite with $n_{\text{quad}} = 40$ converged to ~ 6 significant figures even in the steepest case tested. The observation that the Metropolis–Hastings correction only needs the forward and reverse probabilities *actually used to propose* still holds, and remains the reason that even a genuinely approximate a would not bias the target distribution — but it is not needed to defend this one.

4 Why informedness scales as $\Delta t / \sqrt{T}$

4.1 The derivation

This is the central result of the note, and it is three lines. Inspecting (7), the *only* way Δt , T and M enter $a_{i \rightarrow j}$ is through the standardised gaps $\alpha_{kj} = (\mu_j - \mu_k)/\sigma$. Substituting (2):

$$\begin{aligned} \alpha_{kj} &= \frac{\mu_j - \mu_k}{\sigma} = \frac{-\frac{\Delta t^2}{2M} [(\nabla U)_j - (\nabla U)_k]}{\Delta t \sqrt{T/M}} \\ &= \frac{\Delta t}{2\sqrt{MT}} [(\nabla U)_k - (\nabla U)_j] = \frac{\Delta t g_{kj}}{2\sqrt{MT}}, \quad g_{kj} \equiv (\nabla U)_k - (\nabla U)_j. \end{aligned} \quad (10)$$

The Δt^2 of the drift divides by the Δt of the noise, leaving one power of Δt ; the temperature enters only through σ , as $\sqrt{T}$. Hence:

$p_{\text{match}} \text{ depends on } (\Delta t, T, M, \nabla U) \text{ only through } \frac{\Delta t}{\sqrt{MT}} \text{ and the gradient gaps } g_{kj}. \quad (11)$

4.2 Three consequences

1. An exact invariance. Any two settings with the same $\Delta t/\sqrt{MT}$ have *identical* informedness. With $M = 1$ throughout this project, holding p_{match} fixed while changing temperature requires

$$\Delta t' = \Delta t \sqrt{\frac{T'}{T}}. \quad (12)$$

This is an exact statement about the proposal, not an approximation or a fitted relation.

2. Hotter sampling is less informed at fixed Δt . Raising T inflates σ without touching μ , so the same Δt buys strictly less informedness at higher temperature. There is a certain physical justice to this—a hotter chain should explore more freely—but it means Δt is *not* a transferable constant, and any quoted Δt is meaningless without the T it was measured at.

3. The trap, which we fell into. `test_informedness_vs_dt.py` hard-codes its own $T = 2.0$ as a module constant, so every p_{match} it reports is a $T = 2$ number. Reading $\Delta t = 63$ off that table and using it for a run at $T = 107$ —which is what the first version of the queued command did—changes α by $1/\sqrt{107/2} = 1/7.31$ and lands the proposal essentially back on the uniform floor. In the equivalence of (12),

$$\Delta t = 63 \text{ at } T = 107 \quad \equiv \quad \Delta t = \frac{63}{\sqrt{107/2}} = 8.6 \text{ at } T = 2, \quad (13)$$

and at $\Delta t \approx 8.6$ the measured $T = 2$ curve gives $p_{\text{match}} \approx 0.06$ against a uniform baseline of 0.0526. The correct value for the intended working point is

$$\Delta t = 63 \times \sqrt{107/2} = 63 \times 7.3144 \approx 460. \quad (14)$$

This is now recorded in a comment on `DTS_TO_SWEEP` so that the next reader does not repeat it. For the record, `protein_g`'s long-standing $\Delta t = 1300$ needed no rescaling only because its runs used $T = 2.0$, the same temperature the informedness script assumes—the coupling was invisible for as long as only one temperature was ever used.

5 Choosing Δt

5.1 Why not simply make Δt large?

By (B), large Δt drives $p_{\text{match}} \rightarrow 1$, i.e. a deterministic greedy pick. That is undesirable for three independent reasons.

1. **The gradient's pick is not that good.** By (A), Δt cannot improve it. Measured top-1 accuracy—how often j^* is the true energetic optimum among the 19 competitors, by brute force—is 2/10 on `protein_g` and 10/30 on `zero_polymer` measured away from the energy floor. So a fully greedy proposal would commit to a substitution that is wrong roughly two thirds of the time, with no mechanism left for exploring the alternatives.
2. **A deterministic proposal cripples the Metropolis–Hastings correction.** As $p_{\text{match}} \rightarrow 1$ the proposal becomes a near-deterministic map, the reverse probability $a_{j \rightarrow i}$ for the move that undoes it becomes very small, and the Hastings ratio $a_{j \rightarrow i}/a_{i \rightarrow j}$ swings wildly. The chain's mobility collapses.
3. **Specific sites saturate at wildly different rates.** The measured spread across sites is large—at $\Delta t = 1000$ on `zero_polymer`, p_{match} ranges from 0.40 to 1.00 depending on the site. A Δt chosen for the mean already pins some sites at 1.0.

Equally, Δt too small gives a uniform proposal: correct, but it throws away the gradient information the whole construction exists to exploit, and the Hastings term becomes ≈ 0 because $a_{i \rightarrow j} \approx a_{j \rightarrow i} \approx 1/K_c$.

5.2 The working point

The target is therefore deliberately *partial* informedness—meaningfully above uniform, nowhere near greedy. The convention this project settled on is

$$p_{\text{match}} \approx 0.15, \quad (15)$$

roughly $3\times$ the uniform baseline of $1/19 = 0.0526$ and far from saturation. It leaves the gradient a real but non-binding say, keeps all 19 competitors reachable within a realistic move budget, and keeps both $a_{i \rightarrow j}$ and $a_{j \rightarrow i}$ comfortably away from their extremes so the Hastings correction stays well-conditioned.

5.3 Measured curves

Table 1 is the measured $p_{\text{match}}(\Delta t)$ for both references, both at $T = 2.0$, pooled over 10 brute-forced sites. Note these are *closed-form* evaluations of (7) at the real measured gradients, not Monte Carlo estimates: once ∇U is known at a site (one forward/backward pass), sweeping arbitrarily many Δt values costs nothing further. This is why extending the sweep downward was free.

Δt	protein_g ($L = 56$)		zero_polymer ($L = 566$)	
	mean p_{match}	(min, max)	mean p_{match}	(min, max)
10	—	—	0.0647	(0.056, 0.090)
20	—	—	0.0788	(0.058, 0.137)
50	—	—	0.1285	(0.068, 0.313)
100	—	—	0.2132	(0.084, 0.562)
200	—	—	0.3479	(0.121, 0.818)
300	0.0725	(0.060, 0.099)	0.4436	(0.160, 0.933)
600	0.0954	(0.068, 0.160)	0.6229	(0.275, 0.999)
1000	0.1280	(0.080, 0.256)	0.7447	(0.403, 1.000)
1300	0.1524	(0.089, 0.331)	0.7930	(0.470, 1.000)
2000	0.2047	(0.111, 0.483)	0.8507	(0.503, 1.000)
4000	0.3197	(0.178, 0.743)	0.9156	(0.520, 1.000)
8000	0.4669	(0.300, 0.948)	0.9506	(0.541, 1.000)
16000	0.6266	(0.391, 1.000)	0.9581	(0.581, 1.000)
32000	0.7691	(0.472, 1.000)	0.9659	(0.659, 1.000)

Table 1: Measured $p_{\text{match}}(\Delta t)$, both at $T = 2.0$, $M = 1$, pooled over 10 sites. Uniform baseline $1/19 = 0.0526$. The **protein_g** sweep was never run below $\Delta t = 300$; the **zero_polymer** range was extended down to $\Delta t = 10$ precisely because the old range turned out to sit entirely above its working point.

The two curves are displaced by about a factor of 20 in Δt : at a given Δt , **zero_polymer** is far more informed. By (10) with T and M equal, that displacement is entirely attributable to larger gradient gaps g_{kj} on the longer sequence, which is consistent with the independently measured $\approx 18\times$ larger gradient components there.

target p_{match}	Δt at $T = 2$	Δt at $T = 107$	regime
≈ 0.0526	—	—	uniform floor, no gradient information
0.10	33	240	weakly informed
0.15	63	460	the working point
0.25	127	931	strongly informed
0.50	394	2885	approaching greedy
0.79	1281	9372	effectively greedy

Table 2: Working points for `zero_polymer`, with the $T = 107$ column obtained from the $T = 2$ column by the exact rescaling (12), factor $\sqrt{107/2} = 7.3144$. The $T = 2$ values are linear interpolations within Table 1 and so carry interpolation error; the rescaling itself is exact.

5.4 Recommended practice

1. Never quote Δt without the T it belongs to.
2. Changing T requires rescaling Δt by $\sqrt{T'/T}$, (12), or informedness silently moves.
3. Re-measure Table 1 for a new reference sequence. The curve depends on the gradient gaps, which are a property of the sequence and the energy, and they moved by $20\times$ between these two references.
4. Check the per-site (min, max) spread, not just the mean. A mean of 0.15 can hide sites already saturated at 1.0.

6 What $\Delta t = 460$ vs. $\Delta t = 63$ showed

6.1 The experiment

`validate_boltzmann_toy.py` makes the sampler’s target distribution exactly checkable. It frees exactly two sites (30 and 52), pins all 564 others at the reference, and enumerates every one of the $20^2 = 400$ states by brute force, giving the exact Boltzmann distribution $p_{\text{exact}} \propto \exp(-U/T)$. It then runs the real `_step` logic restricted to those two sites and compares the empirical visit histogram against p_{exact} .

Both runs used $T = 107$ (chosen from the participation-ratio calibration: the inherited $T = 2.0$ puts $p = 1.0000$ on a single state and the chain cannot move), $M = 1$, 20 000 moves, 4000 discarded as burn-in. They differ *only* in Δt , and by §4 that means they differ only in informedness:

run	Δt	informedness at $T = 107$
informed	460	$p_{\text{match}} \approx 0.15$, the intended working point
uniform	63	$p_{\text{match}} \approx 0.06$, essentially the 1/19 floor

The $\Delta t = 63$ run exists because it was the erroneous value in the first version of the queued command. Running it alongside the corrected one turned the mistake into a controlled comparison: the same sampler, same energy, same temperature, with the gradient information effectively switched off.

6.2 Results

6.3 Reading the result

Both pass; this validates the sampler at $L = 566$. The top states come back in essentially the right order and the right proportions, and mean U —the most robust single summary, being

	$\Delta t = 460$ (informed)	$\Delta t = 63$ ($\approx$ uniform)
acceptance rate	0.3060	0.2893
$\text{KL}(p_{\text{emp}} \parallel p_{\text{exact}})$	0.0316	0.0338
total variation distance	0.0787	0.0848
distinct states visited	224/400	223/400
mean U under p_{exact}	+185.20	+185.20
mean U under p_{emp}	+185.63	+187.63
deviation	+0.44 (0.24%)	+2.43 (1.31%)

Table 3: Boltzmann validation on `zero_polymer`, $T = 107$, 20 000 moves, 16 000 post-burn-in samples over 400 enumerated states. $\log 400 = 5.99$ is the rough scale against which to read the KL.

an average over the whole distribution rather than a per-state frequency—agrees to 0.24% at the informed Δt . Neither run shows the decisive failure signature the script warns about: mass piled on states the exact distribution calls unlikely, or the true top states missed. This was the last outstanding safety check and the only one that exercises accept/reject end to end.

The Δt error cost coverage, not correctness. This is the substantive methodological lesson of the pair. A near-uniform proposal still targets the same Boltzmann distribution—the Metropolis–Hastings correction does not care how good the proposal is, only that its forward and reverse probabilities are computed consistently. So the $\Delta t = 63$ run converged perfectly respectably. But with $a_{i \rightarrow j} \approx a_{j \rightarrow i} \approx 1/K_c$, the Hastings term

$$\log a_{j \rightarrow i} - \log a_{i \rightarrow j} \approx 0, \quad (16)$$

which means that run never meaningfully exercised the reverse-probability pass—the second backward pass at the candidate point, and the asymmetry correction it feeds. That is the most intricate part of the sampler and the part most likely to harbour a subtle bug, and it is the reason this test exists. Had only the $\Delta t = 63$ run been done, it would have been reported as a pass while leaving that machinery untested. The informed run is the one that actually tests it.

Informedness helps, but modestly, and not via acceptance. The informed run is better on every metric, but only slightly, and acceptance barely moved (0.306 vs. 0.289). Informedness is not acting as an acceptance-rate lever here. The natural reading is that on a 400-state problem with a ≈ 41 -state effective support, a uniform proposal is already a decent explorer, so there is little headroom; the informed proposal’s advantage should be expected to matter much more on the full 566-site problem, where the competitor set per move is the same 19 but the state space is vastly larger. That expectation is untested.

6.4 Is the residual discrepancy bias or noise?

The KL values are small but not zero, and it is worth being precise about whether that residue is bias. The participation ratio of p_{exact} at $T = 107$ is

$$k_{\text{eff}} = \left(\sum_i p_i^2 \right)^{-1} = 41.4 \quad (17)$$

effective states out of 400. For n *independent* samples from the exact distribution, the expected KL of the empirical histogram is approximately

$$\mathbb{E}[\text{KL}] \approx \frac{k_{\text{eff}} - 1}{2n} = \frac{40.4}{2 \times 16000} = 0.00126, \quad (18)$$

about $25\times$ smaller than observed. But MCMC samples are not independent. Inverting the same expression for the effective sample size actually implied:

$$\text{ESS} \approx \frac{k_{\text{eff}} - 1}{2 \text{KL}} = \begin{cases} 639 & (\Delta t = 460) \\ 598 & (\Delta t = 63) \end{cases} \implies \tau_{\text{auto}} = \frac{16000}{\text{ESS}} \approx 25\text{--}27 \text{ moves.} \quad (19)$$

An autocorrelation time of ≈ 25 moves is entirely ordinary for a two-site chain at 30% acceptance: each site is re-proposed roughly every two moves and the proposal is accepted about 30% of the time, so the state decorrelates over tens of moves. Both KL values are therefore what a *correct* sampler limited by chain autocorrelation looks like. This is an estimate, not a measurement— τ_{auto} was inferred from the KL rather than computed from the chain.

The falsifiable check, and its outcome. If the sampler is unbiased, the KL is pure finite-sample noise and must fall as $1/n$, while the total variation distance—a different functional of the same histogram—must fall as $1/\sqrt{n}$. Predicting forward from the $n = 16000$ run gave $\text{KL}(32000) \approx 0.016$; a KL that instead plateaued near 0.03 would have indicated genuine bias. The run was done (`zero_polymer/validate_boltzmann_results_dt460_M40000.txt`, 40 000 moves, 32 000 post burn-in) and **both laws hold**:

	$n = 16000$	$n = 32000$	ratio	predicted	
$\text{KL}(p_{\text{emp}} \parallel p_{\text{exact}})$	0.0316	0.0169	1.87	2.00 ($1/n$)	pass
total variation distance	0.0787	0.0569	1.38	1.41 ($1/\sqrt{n}$)	pass
τ_{auto} (inferred)	25.0	26.8	—	run-length independent	pass
acceptance rate	0.3060	0.3092	—	unchanged	pass

Table 4: The $1/n$ bias check. KL and total variation are *independent* functionals of the same histogram with different convergence rates, so matching both is considerably stronger evidence than matching either alone: a biased sampler would have each converge to its own nonzero constant rather than keep falling on schedule. The inferred autocorrelation time is a property of the chain and must not depend on run length—it does not.

This settles the question to the resolution available: the residual discrepancy is finite-sample noise, behaving quantitatively as finite-sample noise must, and there is no evidence of bias. The claim can now be stated more strongly than *no detectable bias at this resolution*—though note that no finite experiment can establish exact unbiasedness, and what has been shown is that the error decays on the schedule an unbiased sampler requires.

One number that did not improve, and why that is fine. Mean U moved *away* from the exact value between the two runs ($+0.44 \rightarrow +1.70$). This is not a regression. Mean U is a single scalar statistic carrying its own Monte Carlo error, not a convergence metric: with $\text{ESS} \approx 1200$ and $\text{sd}_p(U)$ of order 150–200, its standard error is roughly 5, so both deviations sit comfortably inside one standard error. The 0.24% agreement at $n = 16000$ reported in Table 3 was therefore partly luck, and should not be read as a precision claim; the KL and total variation, which aggregate over the whole distribution, are the trustworthy convergence diagnostics here.

7 Caveats and open items

1. **The T window is from a two-site subproblem.** The usable range $T \approx 32\text{--}322$, and hence the $\Delta t \approx 460$ derived from it, were obtained with 564 of 566 residues pinned at the reference. A full-length chain explores a far wider energy range, so that window is a starting bracket, not a final answer. `zero_polymer` needs its own temperature sweep.

2. **Informedness curves are floor measurements.** The gradients behind Table 1 were computed at a seq_A only 5 mutations from the reference—0.9% drift at $L = 566$, where the energy sits essentially at its floor. Measured at a comparable drift to `protein_g` ($\approx 8.8\%$), the gradient’s behaviour changes substantially: top-1 rises from 2/10 to 10/30 and the mean true ΔU at the gradient’s pick goes from +474 to -360 . Gradient *gaps*, and hence the whole $p_{\text{match}}(\Delta t)$ curve, should be re-measured in the bulk before Table 2 is trusted for a production run.
3. **The independence approximation** of §2 is the one inexactness in $a_{i \rightarrow j}$. Unquantified against the exact orthant integral.
4. **Per-site saturation** is wide (Table 1), so a single global Δt is necessarily a compromise. Whether a per-site Δt would help, and whether it could be made to preserve detailed balance, is unexplored.
5. **Resolved:** the $1/n$ scaling of the KL (§6.4) was the one outstanding validation item at the time of writing. It has since been run and passed, together with the $1/\sqrt{n}$ scaling of the total variation distance—see Table 4. What remains open on the validation side is only that this is a two-site subproblem: the full 566-site chain has never been validated against anything exact, and cannot be, since its state space is not enumerable.